

Notes: Crawford & Yurukoglu (AER, 2012)

The Welfare Effects of Bundling in Multichannel Television Markets

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1 Big picture

- **Question.** What are the welfare effects of forcing cable and satellite distributors to sell channels a la carte instead of primarily through bundles?
- **Core trade-off.** A la carte gives consumers more choice and reduces payment for unwanted channels, but changes upstream bargaining and can raise per-channel input costs.
- **Main empirical result.** In the baseline full a la carte counterfactual with renegotiation, input costs rise by 103 percent. This largely offsets the consumer benefits of choosing individual channels.
- **Welfare result.** In Table 8, baseline full a la carte with renegotiation raises mean consumer surplus by only 0.2 percent and total surplus by 2.4 percent, despite reducing channels received and watched.
- **Mechanism.** Unbundling makes individual-channel demand more dispersed; this increases channel conglomerates' bargaining stakes and negotiated input costs.
- **Takeaway.** The welfare effect of unbundling cannot be judged from consumer choice alone. It depends critically on input-market bargaining and vertical market structure.

2 Incremental contribution of the paper

- **Structural policy evaluation of bundling.** The paper estimates a full structural model of multichannel television bundling that jointly incorporates consumer demand, channel viewing, distributor pricing, and upstream bargaining.
- **Links demand and usage data.** It combines bundle-purchase data with household-level viewing data, which allows the authors to infer willingness to pay for individual channels.
- **Endogenizes input costs.** Prior a la carte debates often held channel fees fixed. This paper shows that assumption is misleading because unbundling changes the bargaining problem and raises input costs.
- **Full welfare accounting.** The model tracks consumer surplus, distributor profits, channel license-fee revenue, advertising revenue, industry profits, and total surplus.
- **Regulatory contribution.** The paper turns a policy debate into a quantitative counterfactual and shows why retail regulation must consider upstream renegotiation.

3 Data and measurement

3.1 Factbook cable-system data

The Factbook data cover local cable systems from 1997 to 2007. Observations are system-bundle-years and include bundle composition, prices, market shares, homes passed, ownership, and system characteristics.

Interpretation: Factbook data discipline downstream bundle choice and pricing.

3.2 Channel-level and viewing data

SNL Kagan EBCN data provide channel revenue, subscriber quantities, and cost-related variables. Nielsen and MRI data provide channel viewing. MRI household-level viewing data are especially important because bundle purchases alone cannot identify which individual channels households value.

Interpretation: Viewing data allow the model to recover channel-level WTP distributions, including the large mass of households with zero value for many channels.

3.3 Counterfactual outcomes

The paper reports penetration, affiliate fees, consumer expenditure, channels received, channels watched, license-fee revenue, advertising revenue, distributor profits, channel profits, consumer surplus, and total surplus.

Interpretation: The broad outcome set matters because a la carte changes both allocation and surplus division.

4 Models and empirical specifications

4.1 Household subscription and viewing utility

Households choose bundles based on channel values, prices, and outside options:

$$U_{ij} = \text{value of channels in bundle } j - \alpha_i p_j + \text{outside option} + \varepsilon_{ij}.$$

Conditional viewing hours depend on channel tastes and the set of channels available.

Interpretation: Bundle choices identify price sensitivity; viewing choices identify channel-level taste heterogeneity.

4.2 Channel taste distribution

$$Pr(v_{ic} > 0) = \rho_{ic}, \quad v_{ic} \mid v_{ic} > 0 \text{ follows an exponential distribution.}$$

Correlations across channel tastes are estimated to capture substitutability and complementarity across networks.

Interpretation: The model captures the fact that many households never watch a channel while positive viewers have highly skewed viewing hours.

4.3 Nash bargaining for input costs

Input costs are determined by bargaining between distributors and channel conglomerates:

$$\tau_{cd} = \arg \max_{\tau} [\Delta \Pi_d(\tau)]^{\zeta_{cd}} [\Delta \Pi_c(\tau)]^{1-\zeta_{cd}}.$$

Interpretation: The input fee is an equilibrium object. It changes when the retail regime changes from bundling to a la carte.

4.4 Counterfactuals

The model compares bundling to full a la carte, bundle-sized pricing, and theme tiers, with and without renegotiated input costs.

Interpretation: The core policy experiment is general equilibrium within the industry: retail prices and upstream fees are jointly recomputed.

5 Identification and empirical design

- **Demand identification.** Bundle prices, bundle composition, and market shares identify subscription demand.
- **Channel valuation.** MRI and Nielsen viewing data identify individual-channel tastes and taste correlations.
- **Selection correction.** Viewing is observed only for channels a household receives; the model conditions viewing moments on bundle choice.
- **Input-cost identification.** Kagan data and Nash bargaining restrictions identify input-cost parameters and bargaining parameters.
- **Instrument strategy.** The preferred price sensitivity uses prices of the same firm in other markets as instruments.
- **Main threat.** The largest sensitivity comes from the bargaining model, so the paper reports both fixed-input-cost and renegotiated-input-cost counterfactuals.

Interpretation: This is a structural counterfactual design, not a reduced-form causal design. Credibility depends on jointly matching demand, viewing, pricing, and input-cost moments.

6 Tables and figures

6.1 Figure 1: Dispersion in WTP for Components Is Higher than Dispersion in WTP for a Bundle; Figure 2: Nash Bargaining for Input Costs: Bundling versus A La Carte

Read: Figure 1 shows why bundling is profitable: WTP dispersion is smaller for the bundle than for individual components. Figure 2 shows why a la carte can raise input costs through Nash bargaining.

Economic message: These figures provide the conceptual foundation: bundling helps price discrimination, while unbundling changes upstream bargaining stakes.

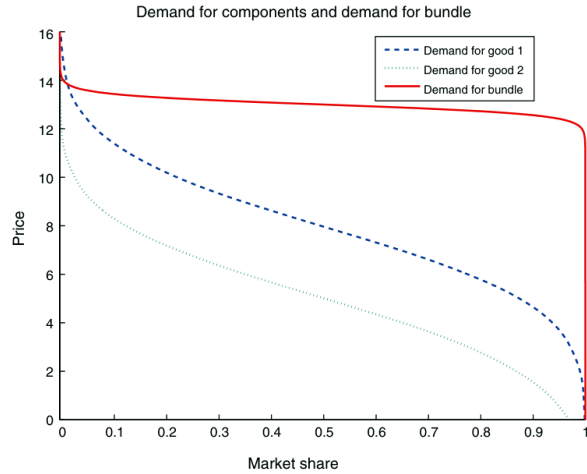


FIGURE 1. DISPERSION IN WTP FOR COMPONENTS IS HIGHER THAN DISPERSION IN WTP FOR A BUNDLE

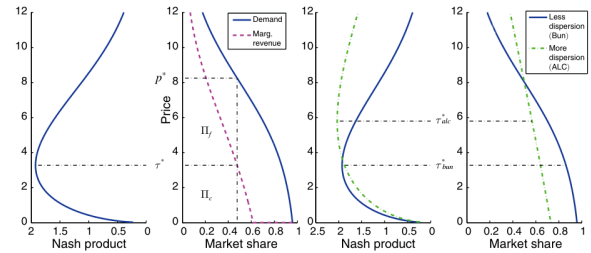


FIGURE 2. NASH BARGAINING FOR INPUT COSTS: BUNDLING VERSUS A LA CARTE

(a) Figure 1: WTP dispersion for components versus bundle.

(b) Figure 2: bargaining under bundling versus a la carte.

6.2 Table 1: Factbook Summary Statistics; Table 2: Channel Summary Statistics

Read: Table 1 summarizes bundle prices, market shares, and numbers of channels. Table 2 summarizes channel carriage and viewing across networks.

Economic message: These tables define the downstream bundle data and the upstream/channel viewing data used in the model.

TABLE 1—FACTBOOK SUMMARY STATISTICS

	N	Mean	SD	Min	Max
All bundles					
Price	25,490	23.46	9.20	0.00	87.06
Market share	25,490	0.44	0.27	0.00	0.99
Total cable channels	25,490	20.3	16.1	0	176
Basic-only markets					
Basic service					
Price	14,732	23.70	6.36	0.00	80.25
Share	14,732	0.54	0.22	0.00	0.99
Total cable channels	14,732	17.3	9.4	0	95
Basic and expanded basic markets					
Basic service					
Price	4,046	13.49	5.71	0.00	47.67
Share	4,046	0.11	0.15	0.00	0.89
Total cable channels	4,046	8.91	7.68	0	56
Expanded basic service					
Price	4,046	27.39	7.92	0.00	87.06
Share	4,046	0.57	0.19	0.00	0.97
Total cable channels	4,046	26.5	10.0	0	77
Basic, exp. basic, and dig. basic markets					
Basic service					
Price	493	13.26	5.60	0.00	38.68
Share	493	0.09	0.09	0.00	0.65
Total cable channels	493	8.3	6.3	1	35
Expanded basic service					
Price	493	34.62	7.81	0.00	61.51
Share	493	0.39	0.16	0.01	0.84
Total cable channels	493	47.1	10.7	18	89
Digital basic service					
Price	493	44.56	10.07	0.00	70.27
Share	493	0.15	0.10	0.00	0.53
Total cable channels	493	78.8	19.1	37	176

Notes: An observation is a system-bundle-year. Prices are in 2000 dollars. Market shares are defined as subscribers divided by homes passed, with homes passed defined as the set of households able to purchase cable service

(a) Table 1: Factbook summary statistics.

TABLE 2—CHANNEL SUMMARY STATISTICS

Data source	Cable system carriage		Household viewership			
	<i>Factbook</i>		Nielsen		MRI	
	Any tier (%)	Basic tier (%)	Mean rating	Mean rating	SD rating	Cume
Channel						
ABC Family Channel	91.2	75.7	0.4	0.6	1.5	31.6
AMC	55.3	30.9	0.5	0.6	1.4	27.2
Animal Planet	22.8	12.1	0.3	0.6	1.5	34.8
Arts & Entertainment	68.3	48.7	0.7	0.8	1.7	37.8
BET Networks	21.1	10.9	0.4	0.3	1.5	10.6
Bravo	13.3	3.3	0.2	0.2	0.7	14.4
Cartoon Network	29.1	15.7	1.6	0.5	1.8	20.9
CNBC	37.6	19.7	0.2	0.5	1.4	29.5
CNN	94.5	77.5	0.7	1.8	3.0	53.8
Comedy Central	25.1	11.1	0.5	0.5	1.3	27.6
Country Music TV	48.0	37.2	0.2	0.2	1.0	13.5
Court TV	16.2	4.5	0.4	0.4	1.4	18.1
Discovery Channel	88.0	71.6	0.6	1.1	1.9	50.9
Disney Channel	41.6	29.6	1.2	0.5	1.4	21.2
E! Entertainment Television	22.9	11.0	0.3	0.3	0.9	24.4
ESPN	96.7	76.7	0.9	1.1	2.2	40.7
ESPN 2	36.6	21.4	0.3	0.5	1.4	25.2
Food Network	13.6	4.5	0.4	0.5	1.5	26.7
Fox News Channel	20.0	10.0	0.8	1.0	2.2	40.0
Fox Sports Net	19.4	11.3	0.3	0.4	1.2	20.2
FX	21.0	9.9	0.5	0.4	1.2	23.3
GSN	8.7	0.8	0.2	0.2	0.9	7.4
Golf Channel	10.9	1.8	0.0	0.1	0.6	6.9
Hallmark Channel	8.2	3.3	0.3	0.2	1.0	10.8
HGTV	26.3	13.2	0.6	0.6	1.6	27.5
History Channel	32.0	18.5	0.6	0.8	1.7	37.9
Lifetime	63.2	41.8	0.9	1.0	2.2	34.4
MSNBC	14.4	5.0	0.3	0.5	1.3	30.2
MTV	52.7	30.2	0.7	0.4	1.4	21.8
MTV2	1.9	0.1	0.0	0.1	0.7	7.8
National Geographic Channel	6.5	1.1	0.1	0.2	0.8	13.2
Nickelodeon	73.8	52.5	1.8	0.4	1.3	17.7
Oxygen	2.8	0.2	0.1	0.1	0.5	7.2
Syfy	33.4	18.4	0.5	0.4	1.4	20.9
SoapNet	4.0	0.4	0.1	0.1	0.6	2.5
Speed Channel	11.8	3.2	0.1	0.1	0.7	7.8
Spike TV	24.0	15.0	0.5	0.4	1.1	18.9
TBS Superstation	96.3	90.7	1.1	0.9	1.7	39.8
The Weather Channel	64.1	46.0	0.3	0.7	1.3	50.3
TLC	45.1	29.9	0.5	0.5	1.3	29.0
TNT	85.2	63.7	1.3	0.9	1.8	41.3
Toon Disney	8.6	2.1	0.2	0.1	0.7	6.1
Travel Channel	16.8	8.3	0.2	0.2	0.7	18.7
TV Guide Channel	19.3	11.5	0.2	0.2	0.6	17.5
TV Land	23.2	15.0	0.8	0.6	1.8	23.9
USA Network	88.8	66.3	1.2	0.8	1.6	37.4
Versus	9.3	1.4	0.1	0.1	0.5	4.8
VH1	39.6	22.6	0.4	0.3	0.9	18.2
WE: Women's Entertainment	7.2	0.8	0.1	0.1	0.5	5.9

Notes: This table reports summary statistics for channels from both our cable system (*Factbook*) and viewership (Nielsen, Mediamark (MRI)) data. The channels reported are those cable channels for which we could get complete data from all three channel data sources used in our analysis. The first column reports the average carriage of each cable channel on any offered tier of service across our system-years. The second column reports average channel carriage on just the basic tier. The last four columns report summary statistics about household viewing patterns across

(b) Table 2: channel summary statistics.

6.3 Figure 3: Distribution of Viewing for CNN, MRI Data; Table 3: Price Sensitivity and Nontelevision Preference Parameters

Read: Figure 3 shows that most households report zero CNN viewing, while positive viewing is highly skewed. Table 3 gives the main price-sensitivity and outside-good parameters.

Economic message: The figure supports the taste distribution assumption; Table 3 supplies the demand parameters used in counterfactual pricing.

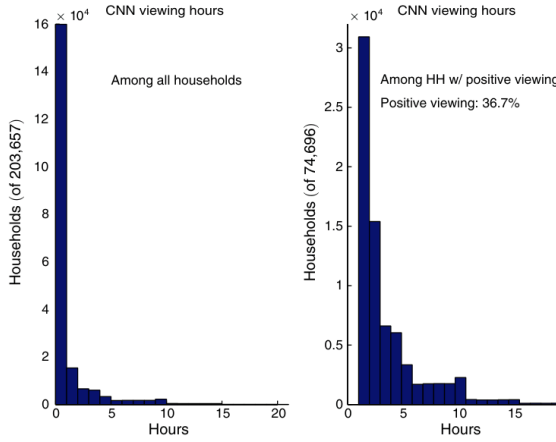


FIGURE 3. DISTRIBUTION OF VIEWING FOR CNN, MRI DATA

Notes: This figure reports the distribution of viewing hours reported by our 200,000+ MRI households. The left panel shows the distribution of viewing for all MRI households, including the 63.3 percent who do not watch CNN. The right panel shows the distribution of viewing among the 36.7 percent of households that do watch CNN.

(a) Figure 3: CNN viewing distribution.

TABLE 3—PRICE SENSITIVITY AND NONTV PREFERENCE PARAMETERS

Parameter	Estimate	Standard error
Price sensitivity (IV)	−0.50	0.03
Price sensitivity (OLS)	−0.29	0.00
Price income interaction	0.11	0.01
Family × outside good	0.00	0.04
Income × outside good	0.64	0.17
Black × outside good	0.70	0.24
Hispanic × outside good	3.97	4.11
Asian × outside good	3.24	1.92
Bachelors × outside good	2.45	0.36
Age × outside good	1.07	0.29

Notes: This table reports our GMM results for a subset of demand parameters, including the estimated mean marginal utility of income, α , the impact of income on marginal utility, π_{yp} , and differences across demographics in tastes for the outside good. Also reported is the estimated mean marginal utility from the same estimation procedure without price instruments, which we denote OLS.

(b) Table 3: demand estimates.

6.4 Table 4: Estimated WTP; Figure 4: Estimated WTP for a Subset of Channels

Read: Table 4 reports mean WTP, dispersion, positive-taste shares, demographic differences, and correlated channels. Figure 4 visualizes WTP distributions.

Economic message: Heterogeneous channel values are the source of the possible consumer benefit from a la carte, but also feed into bargaining.

TABLE 4—ESTIMATED WTP

Channel	Mean WTP	SD WTP	Share positive	Mean WTP family HH	Mean WTP black HH	Highest correlated channel
ABC Family Channel	1.59	2.24	0.49	1.68	1.80	TV Land
AMC	1.40	1.59	0.51	1.15	1.83	MSNBC
Animal Planet	2.05	3.02	0.58	2.08	1.81	National Geographic Channel
Arts & Entertainment	2.10	2.63	0.58	1.90	2.23	History Channel
BET Networks	1.27	2.74	0.34	1.34	4.54	MTV2
Bravo	0.65	0.67	0.61	0.63	0.76	ESPN
Cartoon Network	2.06	4.01	0.49	2.27	2.54	Nickelodeon
CNBC	2.02	2.97	0.55	1.84	2.01	CNN
CNN	5.38	5.91	0.68	4.94	8.30	Fox News Channel
Comedy Central	1.51	2.39	0.61	1.52	1.34	MTV
Country Music TV	0.89	1.56	0.57	0.89	0.79	Food Network
Court TV	1.76	3.11	0.50	1.79	2.23	Arts & Entertainment
Discovery Channel	2.70	2.99	0.65	2.55	2.67	Animal Planet
Disney Channel	1.43	2.51	0.65	1.52	1.72	Nickelodeon
E! Entertainment Television	1.15	1.69	0.62	1.16	1.10	VH1
ESPN	3.08	4.46	0.64	2.86	3.63	ESPN 2
ESPN 2	1.80	3.12	0.62	1.75	2.02	ESPN
Food Network	2.06	3.25	0.71	2.08	2.18	TV Guide Channel
Fox News Channel	4.07	5.89	0.60	4.10	4.69	CNN
Fox Sports Net	1.63	2.82	0.55	1.58	1.55	ESPN 2
FX	1.45	2.59	0.51	1.47	1.41	USA Network
GSN	0.74	2.97	0.08	0.83	1.51	ESPN 2
Golf Channel	0.52	1.86	0.12	0.38	0.68	CNN
Hallmark Channel	1.43	3.96	0.16	1.47	2.09	Country Music TV
HGTV	2.60	4.67	0.42	2.59	3.02	Food Network
History Channel	2.70	4.06	0.40	2.53	3.09	Arts & Entertainment
Lifetime	2.25	3.73	0.31	2.46	5.57	AMC
MSNBC	1.69	3.23	0.29	1.38	2.61	AMC
MTV	1.22	2.28	0.59	1.25	1.36	VH1
MTV2	0.71	1.23	0.52	0.79	0.63	VH1
National Geographic Channel	1.03	1.60	0.69	1.04	0.92	Animal Planet
Nickelodeon	1.31	2.55	0.50	1.45	1.35	Disney Channel
Oxygen	0.41	0.44	0.60	0.49	0.64	Disney Channel
Syfy	1.74	2.97	0.54	1.74	1.82	USA Network
SoapNet	0.49	1.04	0.42	0.52	0.58	TBS Superstation
Speed Channel	0.33	0.41	0.56	0.41	0.19	Versus
Spike TV	1.18	2.00	0.57	1.18	1.07	The Weather Channel
TBS Superstation	2.05	2.85	0.69	1.98	2.23	TNT
The Weather Channel	1.71	1.83	0.70	1.59	1.66	Spike TV
TLC	1.82	2.81	0.61	1.84	1.57	Discovery Channel
TNT	2.36	3.10	0.72	2.31	2.54	USA Network
Toon Disney	0.44	1.69	0.13	0.57	0.90	Cartoon Network
Travel Channel	0.76	2.27	0.15	0.80	0.74	Nickelodeon
TV Guide Channel	0.50	0.75	0.57	0.54	0.60	Food Network
TV Land	2.06	3.40	0.59	2.11	2.45	ABC Family Channel
USA Network	2.12	3.19	0.51	2.19	2.62	TNT
Versus	0.23	0.31	0.49	0.28	0.21	Speed Channel
VH1	0.74	1.28	0.56	0.75	0.90	MTV2
WE: Women's Entertainment	0.45	0.69	0.50	0.49	0.53	National Geographic Channel

Notes: This table reports information of the distribution of WTP for channels implied by our estimates. The first two columns report the mean and standard deviation in WTP for each channel among those that value it positively. The third column reports the estimate share of households that do so. The fourth and fifth columns report estimated WTP among family and black households. The last column reports the channel estimated to have the highest correlation in WTP for each channel. WTP is measured in year 2000 dollars per month per household.

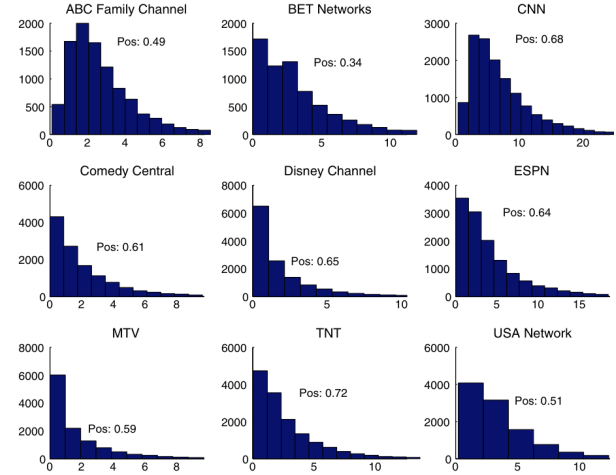


FIGURE 4. ESTIMATED WTP FOR A SUBSET OF CHANNELS

(a) Table 4: estimated WTP.

(b) Figure 4: WTP distributions for selected channels.

6.5 Table 5: Input Cost Parameters; Table 6: Regression Analysis of Distributor Size on Price and Estimated Marginal Cost; Table 7: Conglomerate Bargaining Parameters

Read: Table 5 estimates input-cost shifters. Table 6 shows large distributors have lower prices and marginal costs. Table 7 reports bargaining parameters.

Economic message: These tables identify the upstream input market, which is the key reason a la carte welfare differs from the fixed-cost benchmark.

TABLE 5—INPUT COST PARAMETERS

Parameter	All moments		No bundling moments	
	Estimate	Standard error	Estimate	Standard error
Constant	0.16	0.00	0.20	0.00
Kagan scale	0.91	0.00	0.93	0.00
MSO size	-0.08	0.00	-0.10	0.00
Vertical integration dummy	-0.14	0.01	-0.16	0.01

Notes: This table reports the impact of various factors on our estimated input costs. Kagan scale refers to the input cost for that channel as estimated by Kagan World Media (2008). Distributor (MSO) size is measured in tens of millions of households. Vertical integration is the share of the channel owned by that distributor (between 0 and 1).

(a) Table 5: input-cost parameters.

TABLE 6—REGRESSION ANALYSIS OF DISTRIBUTOR SIZE ON PRICE AND ESTIMATED MARGINAL COST

	Price regression			Estimated marginal cost regression		
	Coeff.	SE	t-statistic	Coeff.	SE	t-statistic
Distributor size	-0.0955	0.0079	-12.12	-0.055	0.0107	-5.10
Number of integrated channels	-0.1668	0.0684	-2.44	0.473	0.093	5.07
Dummy variables						
Channels	Yes			Yes		
Year	Yes			Yes		
Tier	Yes			Yes		
Number of bundles	Yes			Yes		
Year × tier	Yes			Yes		
Number of bundles × tier	Yes			Yes		
N	25,490			25,490		
R ²	0.563			0.169		
F(271, 25,218)	111.92			18.98		

Notes: This table reports the results of regressions designed to highlight the identification of our input cost estimates. The first set of columns reports the results of a regression of bundle prices on the size of the distributor offering the bundle and a sum of the number of vertically integrated channels in the distributor's bundle. We condition

(b) Table 6: distributor size and marginal cost.

TABLE 7—CONGLOMERATE BARGAINING PARAMETERS

Conglomerate	Big cable	Small cable	DirecTV	Dish Network
ABC Disney	0.28	0.25	0.18	0.17
Viacom	0.49	0.48	0.54	0.53
NBC Universal	0.50	0.49	0.52	0.51
Comcast (Content Division)	0.69	0.68	0.67	0.66
Scripps	0.55	0.55	0.58	0.58
News Corporation	0.42	0.39	0.34	0.32
Rainbow Media	0.70	0.69	0.68	0.67
Discovery Networks	0.62	0.61	0.63	0.63
Time Warner	0.40	0.38	0.38	0.37
Hallmark	0.69	0.69	0.71	0.71
Lifetime	0.43	0.43	0.43	0.43
Oxygen	0.73	0.72	0.71	0.70
Weather Channel	0.69	0.69	0.69	0.69
TV Guide	0.77	0.77	0.76	0.76

Notes: This table reports our estimated bargaining parameters for channel conglomerates versus distributors of various types. Smaller values of the bargaining parameters indicate relatively more bargaining power for channels. Channel conglomerates are ABC Disney (ABC Family Channel, Disney Channel, ESPN, ESPN2, Soap Net, Toon Disney), Viacom (BET Networks, Comedy Central, Country Music TV, GSN, MTV, MTV2, Nickelodeon, Spike TV, TV Land, VH1), NBC Universal (Arts and Entertainment, Bravo, CNBC, MSNBC, Syfy, USA Network), Comcast (E! Entertainment Television, Golf Channel, Versus), Scripps (Food Network, HGTV), News Corporation (Fox News Channel, Fox Sports Net, FX, National Geographic Channel, Speed Channel), Rainbow Media (AMC, WE: Women's Entertainment), Discovery Networks (Animal Planet, Discovery Channel, History Channel, TLC, Travel Channel), Time Warner (Cartoon Network, CNN, Court TV, TBS Superstation, TNT). Hallmark, Lifetime, Oxygen, Weather Channel, and TV Guide are single-channel "conglomerates." See the end of Section IV for descriptions of the distributor types.

Table 7: conglomerate bargaining parameters.

6.6 Table 8: Baseline Counterfactual Results: Full A La Carte; Table 9: Input Cost and Welfare Effects by Channel

Read: Table 8 is the main welfare table. Without renegotiation, consumer surplus rises by 19.2 percent; with renegotiation, consumer surplus rises by only 0.2 percent. Table 9 shows channel-level cost and profit effects.

Economic message: The central conclusion is that input-cost renegotiation can erase the simple consumer-choice gains from a la carte.

	Bundling	ALC no renege	Percent change	ALC with renege	Percent change
Nonwelfare outcomes					
Cable & sat penetration	0.880	0.998	13.3	0.993	12.8
Total affiliate fees	\$18.22	\$18.22	0.0	\$36.98	103.0
Mean consumer expn	\$27.63	\$21.07	−23.8	\$28.24	2.2
Number channels received	42.8	22.0	−48.5	19.3	−54.9
Number channels watched	22.2	22.0	−0.5	19.3	−12.8
Welfare outcomes					
Channel profits					
Total license fee rev	\$16.03	\$7.95	−50.4	\$15.44	−3.7
Total advertising rev	\$13.38	\$14.71	10.0	\$14.73	10.1
Total channel revenue	\$29.41	\$22.67	−22.9	\$30.16	2.6
Distributor profits	\$11.59	\$13.11	13.1	\$12.81	10.4
Total industry profits	\$41.00	\$35.78	−12.7	\$42.97	4.8
Mean consumers surplus	\$45.82	\$54.59	19.2	\$45.91	0.2
Total surplus	\$86.82	\$90.37	4.1	\$88.88	2.4

Notes: This table reports the results of our baseline counterfactual simulations of full à la carte (ALC) pricing policies on prices and welfare. The economic environment consists of one large and one small cable market (served by one large and one small cable operator) and two “national” satellite providers, each offering access to their platform and approximately 50 cable channels. In the bundling equilibria reported in column one, each firm competes by pricing a single bundle of channels. In both ALC equilibria, each firm competes by setting a fixed fee and then separate prices for each offered channel. Columns two and three report results for ALC *without* allowing input-market renegotiation (i.e., with input costs at their values in the bundling equilibrium); columns four and five allow renegotiation. In the renegotiation equilibrium, we impose that downstream prices equal the renegotiated input costs. See footnote 39 in the text for details. Average outcomes (e.g., Total Affiliate Fees, Number of Channels) are

(a) Table 8: baseline full a la carte counterfactual.

Channel	Input cost effects			Profit effects				
	Bundling input cost	ALC input cost	Percent change	Total bundling revenue	Total ALC revenue	Percent change	Percent change license fee rev.	Percent change advert. rev.
ABC Family Channel	\$0.32	\$0.83	156.9	\$0.46	\$0.58	24.5	29.9	15.9
AMC	\$0.32	\$0.54	67.8	\$0.41	\$0.43	3.9	−2.2	16.9
Animal Planet	\$0.20	\$0.97	372.8	\$0.25	\$0.53	109.3	150.0	9.8
Arts & Entertainment	\$0.31	\$1.08	250.6	\$0.57	\$0.91	58.8	109.4	13.3
BET Networks	\$0.26	\$0.58	127.3	\$0.56	\$0.55	−1.7	−26.8	15.4
Bravo	\$0.27	\$0.51	92.3	\$0.39	\$0.40	1.4	2.0	0.6
Cartoon Network	\$0.26	\$0.78	199.1	\$0.54	\$0.62	14.7	19.4	11.3
CNBC	\$0.34	\$0.93	170.6	\$0.53	\$0.70	30.7	43.7	13.6
CNN	\$0.49	\$2.92	498.0	\$0.81	\$1.98	144.1	265.3	7.2
Comedy Central	\$0.23	\$0.66	187.5	\$0.61	\$0.72	18.2	43.2	5.8
Country Music TV	\$0.18	\$0.56	211.1	\$0.26	\$0.29	10.8	17.7	0.2
Court TV	\$0.22	\$0.85	276.1	\$0.35	\$0.49	41.5	63.9	12.2
Discovery Channel	\$0.34	\$1.47	339.6	\$0.59	\$1.16	95.9	182.0	10.0
Disney Channel	\$0.77	\$0.70	−8.9	\$0.68	\$0.27	−59.6	−59.6	0.0
E! Entertainment Television	\$0.30	\$0.48	62.0	\$0.41	\$0.38	−7.6	−15.8	7.2
ESPN	\$2.44	\$0.87	−64.5	\$3.80	\$2.33	−38.6	−75.9	9.5
ESPN 2	\$0.33	\$0.71	114.2	\$0.46	\$0.48	3.9	1.8	7.7
Food Network	\$0.19	\$0.85	352.9	\$0.49	\$0.71	44.0	122.1	4.5
Fox News Channel	\$0.36	\$1.83	411.8	\$0.70	\$1.27	82.4	171.8	8.9
Fox Sports Net	\$1.56	\$0.79	−49.3	\$1.51	\$0.46	−69.4	−77.4	8.9
FX	\$0.36	\$0.68	90.3	\$0.61	\$0.58	−5.3	−19.8	10.2
GSN	\$0.19	\$0.42	124.3	\$0.23	\$0.12	−47.7	−76.0	20.7
Golf Channel	\$0.32	\$0.14	−57.5	\$0.37	\$0.10	−72.6	−99.9	14.9
Hallmark Channel	\$0.17	\$0.63	272.5	\$0.33	\$0.32	−3.7	−28.6	17.1
HGTV	\$0.25	\$1.04	310.8	\$0.60	\$0.82	38.4	77.2	15.2
History Channel	\$0.29	\$2.29	699.5	\$0.53	\$1.16	120.5	237.0	13.5
Lifetime	\$0.32	\$0.85	166.8	\$0.81	\$0.88	9.3	−4.6	16.7
MSNBC	\$0.26	\$0.69	168.3	\$0.33	\$0.31	−4.8	−14.6	16.1
MTV	\$0.37	\$0.47	28.3	\$1.02	\$0.93	−8.4	−44.6	8.6
MTV2	\$0.17	\$0.54	223.0	\$0.19	\$0.21	9.4	12.4	−0.5
National Geographic Channel	\$0.29	\$0.65	120.9	\$0.34	\$0.32	−5.1	−6.2	−1.2
Nickelodeon	\$0.48	\$0.45	−7.5	\$1.38	\$1.23	−10.5	−61.8	12.5
Oxygen	\$0.24	\$0.09	−63.7	\$0.31	\$0.16	−48.0	−76.1	16.5
Syfy	\$0.27	\$0.70	160.0	\$0.55	\$0.63	15.3	18.3	13.0
SoapNet	\$0.22	\$0.44	98.8	\$0.24	\$0.15	−37.9	−47.0	3.7
Speed Channel	\$0.27	\$0.42	56.7	\$0.32	\$0.18	−43.9	−51.8	−21.3
Spike TV	\$0.29	\$0.60	106.7	\$0.54	\$0.53	−1.1	−8.6	5.8
TBS Superstation	\$0.38	\$0.88	132.0	\$0.89	\$1.04	16.5	33.1	6.6
The Weather Channel	\$0.22	\$0.60	174.4	\$0.34	\$0.56	64.7	102.4	15.1
TLC	\$0.27	\$0.83	205.9	\$0.42	\$0.57	35.7	55.5	9.5
TNT	\$0.84	\$0.93	11.1	\$1.35	\$1.15	−15.2	−33.6	6.9
Toon Disney	\$0.21	\$0.39	86.1	\$0.24	\$0.10	−57.9	−83.2	17.7
Travel Channel	\$0.26	\$0.45	69.7	\$0.32	\$0.16	−50.5	−74.9	14.4
TV Guide Channel	\$0.16	\$0.14	−16.2	\$0.24	\$0.18	−24.3	−49.4	15.9
TV Land	\$0.21	\$0.86	301.1	\$0.34	\$0.53	57.0	92.8	11.9
USA Network	\$0.51	\$0.84	65.0	\$1.13	\$1.17	3.7	−12.2	14.1
Versus	\$0.25	\$0.29	17.7	\$0.26	\$0.13	−51.8	−60.4	−8.9
VH1	\$0.24	\$0.44	80.8	\$0.55	\$0.50	−9.7	−27.3	1.4
WE: Women's Entertainment	\$0.22	\$0.32	46.1	\$0.26	\$0.19	−28.5	−39.8	5.1
Total	\$18.22	\$36.98	103.0	\$29.41	\$30.16	2.6	−3.7	10.1

Notes: This table reports the results by channel of the input cost and profit consequences from our baseline. Full à la carte (ALC), counterfactual with input-cost renegotiation. As in Table 8, downstream prices are set at the renegotiated input costs; see footnote 39 for details. The first three columns report weighted averages (across distributors) of our estimated per-subscriber input costs under bundling and ALC equilibria (and their associated change). They are measured in 2000 dollars per subscriber per month. Distributors must pay the bundle-input cost for all their subscribers in the bundling counterfactual, but pay the ALC input cost only for those that choose to subscribe under the ALC counterfactual. The remaining columns summarize the profit effects by channel. The fourth through

(b) Table 9: channel-level input-cost and welfare effects.

6.7 Table 10: Alternative Counterfactual: Full ALC, Bundle-Sized Pricing, and Theme Tiers; Figure A1: Television Programming Industry

Read: Table 10 studies alternative unbundling policies. Figure A1 maps the content, distribution, and advertising markets in the television programming industry.

Economic message: Policy design matters, but the broad conclusion remains: retail policy must account for vertical bargaining and advertising markets.

TABLE 10—ALTERNATIVE COUNTERFACTUAL: FULL ALC, BUNDLE-SIZED PRICING, AND THEME TIERS

	Levels				Percent change		
	Bundling	Full ALC	Bundle-sized pricing	Theme tiers	Full ALC	Bundle-sized pricing	Theme tiers
Nonwelfare outcomes							
Cable & sat penetration	0.880	0.993	0.987	0.977	12.8	12.1	11.0
Total affiliate fees	\$18.22	\$36.98	\$34.44	\$35.49	103.0	89.1	94.9
Mean consumer expn	\$27.63	\$28.24	\$28.60	\$36.98	2.2	3.5	33.8
Number channels received	42.8	19.3	17.0	34.7	-54.9	-60.3	-18.8
Number channels watched	22.2	19.3	15.8	19.2	-12.8	-28.7	-13.4
Welfare outcomes							
Channel profits							
Total license fee rev	\$16.03	\$15.44	\$17.97	\$25.26	-3.7	12.0	57.5
Total advertising rev	\$13.38	\$14.73	\$14.44	\$13.95	10.1	7.9	4.3
Total channel revenue	\$29.41	\$30.16	\$32.40	\$39.20	2.6	10.2	33.3
Distributor profits	\$11.59	\$12.81	\$10.63	\$11.72	10.4	-8.3	1.1
Industry profits							
Total industry profits	\$41.00	\$42.97	\$43.03	\$50.93	4.8	5.0	24.2
Mean consumers surplus	\$45.82	\$45.91	\$41.79	\$35.73	0.2	-8.8	-22.0
Total surplus	\$86.82	\$88.88	\$84.82	\$86.66	2.4	-2.3	-0.2

Notes: This table reports the results of alternative counterfactual simulations of various policy interventions on prices and welfare. The economic environment is as in Table 8. Columns one, two, and five report the counterfactual outcomes in bundling and full à la carte (ALC) environments as in Table 8. The remaining columns report counterfactual outcomes under Bundle-Sized Pricing and Theme Tiers. In the Bundle-Sized Pricing counterfactual, each downstream distributor competes by offering a full bundle of all the channels and a second bundle of 15 channels, the identities of which may be chosen by each household. In the Theme Tier counterfactual, each downstream distributor competes by setting a fixed fee and offering five theme tiers from which the household can choose any combination. The theme tiers are Sports (ESPN, ESPN 2, Fox Sports Net, Golf Channel, Speed Channel, Versus), News (CNBC, CNN, Fox News Channel, MSNBC), Family and Education (ABC Family Channel, Animal Planet, Discovery Channel, Disney Channel, History Channel, National Geographic Channel, Nickelodeon, TLC, Toon Disney), Music and Lifestyle (Bravo, Country Music TV, E! Entertainment Television, Food Network, HGTV, Lifetime, MTV, MTV2, Oxygen, SoapNet, TV Guide Channel, VH1, WE: Women's Entertainment), and General (AMC, Arts and Entertainment, RFT Networks, Cartoon Network, Comedy Central, Court TV, FX, GSN, Hallmark Channel, Sci-Fi, Spike TV, TRS).

(a) Table 10: alternative counterfactuals.

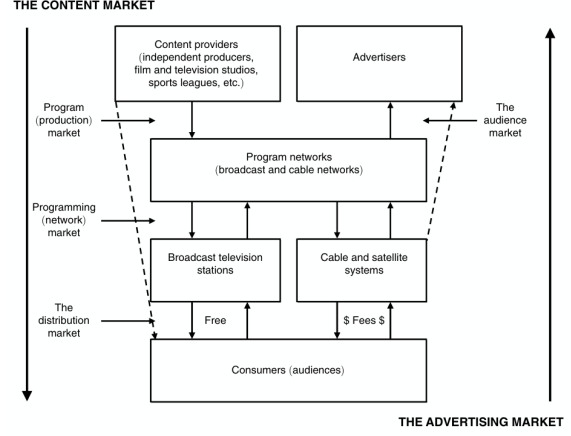


FIGURE A1. TELEVISION PROGRAMMING INDUSTRY

(b) Figure A1: television programming industry.

7 Short summary

- The paper estimates a structural model of the multichannel television industry.
- It combines bundle purchase data, viewing data, channel revenue/cost data, and Nash bargaining.
- A la carte expands consumer choice but changes upstream bargaining.
- Input costs rise sharply under full a la carte with renegotiation.
- Without renegotiation, full a la carte raises consumer surplus; with renegotiation, gains are nearly eliminated.
- Alternative pricing systems do not overturn the importance of input-cost renegotiation.

8 Conclusion

8.1 Core conclusion

Crawford and Yurukoglu show that a la carte cable pricing has ambiguous welfare effects once the full vertical structure is modeled. Consumer choice expands, but input costs rise.

8.2 Economic intuition

Bundling compresses WTP dispersion and helps downstream price discrimination. Unbundling reveals individual channel valuations and changes the bargaining surplus between distributors and channels.

8.3 Incremental contribution

The paper's incremental contribution is to endogenize upstream input costs in a structural policy evaluation of downstream unbundling.

8.4 Policy implication

Mandatory a la carte is not automatically consumer friendly. Holding input costs fixed can dramatically overstate consumer gains.

8.5 Takeaway

Bundling policy must be evaluated as a vertical-market problem, not just a retail consumer-choice problem.